The Non-Specificity of Placebo Effects

Abstract

The notion of placebo effect plays an important role in medicine, yet there is no widely accepted characterization of it. This paper defends and refines the view, originating with psychiatrist Arthur Shapiro, that placebo effects are characterized by their non-specificity. Although once influential, Shapiro's view has faced strong criticism and has largely fallen out of favor. I argue, however, that there are two important and defensible senses in which placebo effects are non-specific, both of which can be made precise within Woodward's interventionist framework of causation. First, placebo effects lack one–one specificity: when a treatment produces a response via the placebo effect, countless other treatments could produce the same outcome, and the treatment could easily produce many different outcomes. Second, the outcome of a placebo effect cannot be attributed specifically to the treatment itself; instead, patient expectations and conditioning history are the more important causes of the outcome. Drawing on the interventionist account of the dimensions that influence our judgments of causal importance, I show that placebos count as mere background conditions of the responses they provoke because of the non-specificity of their effects and their failure to meet closely related interventionist conditions of proportionality and stability. Properly articulated, the non-specificity account of placebo effects offers substantial conceptual advantages and fares better than competing views in multiple respects.

Keywords: Placebo effect; placebo; specificity; interventionism; causal models; psychotherapy

The placebo effect lies at the center of several important issues in medicine, including the ethics of deception in clinical practice, the status of alternative medicine, and the methodology of randomized controlled trials. Yet it is notoriously hard to define what placebo effects are, and the notion gives rise to conceptual puzzles. In particular, speaking of placebo *effects* implies that placebos cause therapeutic responses, but the notion of "placebo" itself strongly connotes causal impotence. This tension is evident in H.K. Beecher's paper "The Powerful Placebo" (1955), which launched the field of placebo studies, where placebo effects are defined as "real therapeutic effects" of "pharmacologically inert substances" – a formulation that borders on logical contradiction (Howick 2017). More recent attempts at definition also face difficulties, and none has achieved consensus. Indeed, several authors have argued that the concept of placebo effect is fundamentally confused and should be abandoned altogether (Gøtzsche, 1994; Nunn, 2009; Turner, 2012).

This paper argues that we can make good sense of the placebo effect by reviving an old account due to the psychiatrist Arthur Shapiro, according to which the mark of placebo effects is their *non-specificity* (Shapiro, 1968; Shapiro & Morris, 1978). Once commonly accepted in the medical community, this view has been strongly criticized by medical researchers and philosophers (e.g. Grünbaum, 1986; Kirsch, 2005; Nunn, 2009; Howick, 2017; Friesen, 2020) and has now largely fallen out of favor. In essence, critics have alleged that Shapiro's account is intolerably vague, and that attempts to make it more precise fall prey to decisive objections. However, I will argue that placebo effects can accurately be characterized as non-specific in two distinct but related ways, each of which plausibly captures at least part of what Shapiro had in mind. First, when a placebo effect takes place, treatment and therapeutic response do not map one-to-one onto each other: any number of alternative treatments could produce the same response, and the very same treatment could easily elicit different responses. While 'active' (non-placebic) effects are often not strictly one-to-one specific either, this lack of specificity is far more extreme in the case of placebo effects. Second, placebo effects are non-specific in the sense that the treatment is not the main cause of the response it provokes. On this second interpretation, the non-specificity of placebo effects is tied to causal selection, the fact that among all the factors on which a given outcome depends, we generally single out some as the main (or 'real') causes of that outcome and others as mere background conditions for the outcome's occurrence. The thought here is that in an instance of placebo effect, the most important causal determinants of the patient's response are features of the patient themselves (their expectations or conditioning history), while the treatment itself is a mere background condition. Thus, the response cannot be attributed *specifically* to the treatment; rather, the chief cause lies elsewhere. Though these readings have been considered and rejected in the critical literature on Shapiro, I will argue that properly spelled out they do provide a plausible criterion for distinguishing placebo effects from active medical effects. To articulate and defend those two readings of the non-specificity thesis, I will draw on Woodward's interventionist account of causation, including his account of causal specificity and related notions as well as related work on causal selection (Woodward 2010, 2021).

As I will show, when put on proper footing, the non-specificity account of placebo effects brings significant conceptual benefits. It avoids the problems faced by other major attempts to define the placebo effect and answers important questions that they leave open. It explains and justifies the common intuition that despite having real medical effects, placebos are second-rate treatments inferior to 'real' treatments. It also elucidates puzzling features of the placebo effect, e.g. the fact that placebos are naturally described as inert despite being capable of causing real therapeutic responses. Overall, against appeals to eliminate the notion of placebo effect, the non-specificity account shows that this notion is a consistent one that has a legitimate role to play in medical theory and practice.

1. Leading accounts of the placebo effect

The link between the notions of placebo effect and of placebo is subtle (Howick, 2017, p. 1370). The administration of a placebo need not induce a placebo effect in the patient, and an 'active' (i.e., non-placebo) treatment may well produce its response partly through a placebo effect. Following Grünbaum (1986), we can clarify these matters by defining active and placebo treatments in terms of the placebo effect, as follows. A medical intervention – for example, the administration of a substance, or a surgical act – is a *placebo treatment* with respect to a medical condition *D* just in case its effect on *D* (if any) is entirely a placebo effect.[1] (When the medical intervention consists in the administration of a substance, then the substance is a placebo.) By contrast, a medical intervention is an *active treatment* for *D* when its effect on *D* is not, or not solely, due to the placebo effect. (The non-placebic portion of the effect is the *active effect* of the treatment.) These definitions acknowledge the insight, also due to Grünbaum (1986), that whether a treatment is placebic depends on the targeted condition: for example, taking a sugar pill is a placebo treatment for pain relief, but an active treatment for hypoglycemia.[2] Note also that on these definitions, placebo treatments are necessarily *medical* acts. As Rees (2024) points out, a qualification of this sort is required to distinguish placebos from ordinary occurrences in which positive feedback improves well-being. Following Rees, we can understand medical acts broadly to include contexts in which the patient takes themselves to be subjected to a medical

[1] If the effect is negative (i.e., the treatment produces or worsens *D*) we have a case of the 'nocebo effect' (Howick, 2021; Due, 2025).

[2] This helps explain the phenomenon of 'active placebos' (see Holman, 2015), i.e. substances that mimic the side-effects of a drug (e.g. nausea) but are not active treatments for the condition treated by the drug. In this case we can say that the substance is a placebo for the condition treated but an active (harmful) treatment for the medical conditions produced as side-effects.

act, even if that belief is false, e.g. if the patient is being sold snake oil by a con-artist.[3] What remains, then, is to explain what a placebo effect is. Before turning to Shapiro's account, it will be helpful to briefly review three leading attempts to do so.

Among medical researchers, a popular view is that of Kirsch (2005). To motivate it, consider a classic example of the placebo effect: the experiment of McGlashan *et al*. (1969), in which participants experienced significant pain relief when given a pill falsely presented by the experimenters as a powerful pain-killer. Here the therapeutic response was produced through a psychological mechanism, namely through patient expectations of treatment efficacy. Kirsch therefore proposes to define the placebo effect as "that portion of the treatment effect that was produced psychologically, rather than through physical means" (2005, p. 791). But this account faces severe problems (Friesen, 2020). It presupposes an untenable form of mind-body dualism, and ignores the fact that placebo treatments produce real physiological responses, just like active treatments do. (For example, placebo painkillers reduce pain by stimulating the release of endogenous opioids into the bloodstream (Colloca & Benedetti, 2005).) In addition, expectations are not the only mechanism underlying placebo effects. Another one is conditioning: a stimulus such as flavor or color is first paired with a substance or procedure that produces a given physiological response, and later induces that response alone. For instance, in a study by Giang *et al*. (1996), multiple sclerosis patients were given an anise-flavored syrup along with cyclophosphamide (an immunosuppressant that inhibits lymphocyte production), and later displayed lymphocyte decrease when given the syrup alone. This is commonly regarded as an instance of placebo effect, but whether the relevant mechanism counts as "psychological" is highly dubious. Finally, Kirsch's account also implies that all psychotherapies are placebo treatments. But even if that were so, this should be established on empirical and not conceptual grounds (Howick, 2017).

Among philosophers, the most influential account is probably Grünbaum's (1986), which defines the placebo effect as the portion of treatment effect produced by an "incidental" feature of a treatment, i.e. any feature that the theory behind the treatment regards as causally irrelevant to patient response. For example, if the taste of a pill is deemed therapeutically irrelevant, but actually induces the desired patient response, we have a placebo effect. This account has

[3] We can also count experiments designed to test a medical treatment as medical acts, which allows us to include both the use of placebo treatments in clinical practice and in clinical trials.

generated a number of criticisms (Greenwood, 1997; Waring, 2003), some of which can be addressed by suitable refinements (Howick 2017). But key problems remain. A crucial issue is that therapeutic theories and hence what counts as a placebo may vary from practitioner to practitioner. While Grünbaum embraced this relativity, one may worry that this makes his account too uninformative (Friesen 2020, p. 13). The account also has strange implications. For example, in McGlashan et al.'s (1966) study, the experimenters themselves theorized (rightly) that patient response was induced by the treatment's resemblance to a real drug. On Grünbaum's framework, the effect therefore does not qualify as placebic, at least not relative to the right theory of the treatment. Yet this is clearly a paradigmatic case of the placebo effect.[4]

Finally, Friesen (2020) proposes to define the placebo effect in terms of its underlying mechanisms, viz. as "any positive clinical outcome that is brought about via expectations or conditioning" (2020, 14). While this serves as a useful working definition, it leaves important questions unresolved. For instance, as Friesen herself acknowledges, it remains an open question in placebo studies whether mechanisms beyond expectancy and conditioning might also generate placebo effects (see e.g. Moerman 2002). Accordingly, Friesen allows that her definition may need to be revised if evidence of additional mechanisms emerges. Yet her account does not clarify when and why it would be legitimate to classify a newly discovered therapeutic mechanism as producing a placebo effect. Nor does it explain what expectancy and conditioning have in common that justifies grouping them together under a single medical category and distinguishing them from active effects of treatments. Thus, despite being intended as a response to eliminativist concerns, her account does not fully dispel doubts about the legitimacy and usefulness of the placebo/non-placebo distinction. As we will see, when suitably elaborated, the non-specificity account of placebo effects can address these problems.

2. Placebo effects as lacking one-one specificity

The thesis that the defining feature of placebo effects lies in their "non-specificity" was first advanced by the psychiatrist Arthur Shapiro. In his (1968), he defined the placebo effect as "the non-specific psychologic or psychophysiologic effect produced by a placebo" (1968: 682). This neglects the fact that active treatments can produce placebo effects, but this is easily remedied –

[4] See Rees (2020, 242) for a related criticism.

we can simply define the placebo effect as the portion of the total treatment effect that is non-specific. A much deeper problem is that despite the obvious need for clarification, the paper offered no account of what “non-specific” means. A decade later, Shapiro refined the definition by stating that “specific activity is the therapeutic influence attributable solely to the contents or processes of therapies rendered” (Shapiro & Morris 1978: 372). But as Grünbaum (1986) rightly observes, without further explanation this formulation is also rather cryptic. A possible reading of Shapiro’s 1968 statement, suggested by the word “psychologic”, is that it treats the placebo effect as a generic form of anxiety relief or mood improvement without effect on the particular condition treated. But so understood, Shapiro’s definition is clearly incorrect: as Friesen (2020: 9) notes, placebo effects have been demonstrated for a large number of specific conditions, including pain, asthma attacks, migraines, itches, impotence, rhinitis, memorization, multiple sclerosis, and arm rigidity in Parkinson patients. More broadly, critics have argued that there is no simply no way of spelling out ‘non-specificity’ that yields an empirically satisfactory account of the placebo effect. These criticisms have largely prevailed: while Shapiro’s account was once widely accepted among placebo researchers, it has now fallen out of favor. But there are in fact two plausible ways to interpret ‘specificity’ (and lack thereof) on which Shapiro’s thesis furnishes an empirically plausible and conceptually attractive account of placebo effects. Critics have considered these interpretations, but their dismissal has been too hasty, as we will see.

Woodward’s (2010) causal account of specificity in the biomedical sciences supplies the first relevant interpretation. That account relies on a minimal interventionist criterion of causation, on which a variable X is a cause of variable Y when there exist background circumstances in which some intervention on X would change Y’s value.[5] On that basis, Woodward identifies two forms of specificity that causal relationships can have and that are of special significance in biology and medicine. One is ‘fine-grained influence’, which concerns the extent to which the state of the cause can be finely modulated by intervention so as to influence the precise state of the effect. A paradigm example is the fact that a sequence of DNA can be altered in many ways that each lead to a different protein being synthesized. The second form of specificity, and the relevant one for our purposes, is one-one specificity. In a nutshell, the relationship between a cause X and an effect Y is one-to-one specific insofar as X and Y are

[5] See Woodward (2003, 98) for a definition of interventions.

uniquely associated with each other. It is useful to distinguish two dimensions of one-one specificity. The first is the extent to which the cause produces only the effect under consideration, or many other effects as well (within a contextually specified range). The second dimension is the extent to which the effect is associated uniquely with the cause, or can be produced by many other causes instead (again, within a contextually specified range). I will call the first dimension ‘specificity of effect’ and the second ‘specificity of cause’. As an example, consider the ‘lock and key’ model of enzyme action (Woodward 2010: 309). On that model, the causal relationship between a given enzyme and substrate is fully one-one specific as the enzyme can bind only that substrate (specificity of effect), and only that enzyme can bind the substrate (specificity of cause). Other significant forms of one-one specificity in the biomedical sciences include antibody specificity (Woodward 2010: 312), ligand-protein binding specificity (Lean, 2020), host specificity in ecology (Blanchard, 2022a) and specificity of association in epidemiology (Blanchard, 2022b).

This notion of one-one specificity, I claim, supplies a plausible interpretation of Shapiro’s thesis: placebo effects are non-specific in that the causal relationship between the treatment and the response exhibits a massive lack of one-one specificity. When a treatment induces a response via the placebo effect, any number of other treatments could easily have produced the same response (provided the patient had the right expectations or conditioning history), and the treatment could easily have produced a different response instead (provided again, that the patient had the suitable expectations or conditioned dispositions.) So the treatment-response relationship lacks both specificity of effect and specificity of cause. This reading of the non-specificity thesis likely captures at least part of what Shapiro had in mind. His claim that “specific activity is the therapeutic influence attributable solely to the contents” of treatments is naturally read as stating that an effect is specific to the extent that no or few other treatments could produce the same one, and hence as stating that lack of specificity of cause is a distinguishing feature of placebo effects (though as we will see lack of specificity of effect is another important trait).

To illustrate, consider a patient who experiences pain relief after taking a sugar pill presented as a powerful analgesic, as in McGlashan *et al.*’s (1969) study.[6] Clearly, the patient

[6] Though their paper actually does not specify the content of the placebo pill they used.

would have responded the same way if they had been given some other treatment - a lactose pill, a topical cream, a cataplasm or what have you – so long as the treatment was still presented as an effective painkiller. More generally, when a placebo effect is generated through patient expectations, virtually any treatment could have produced the same outcome, provided the expectation of efficacy remained intact. The same holds for placebo effects arising from conditioning. Classic cases involve a placebo that mimics the color, shape, or taste of a drug familiar to the patient, thereby eliciting the same physiological response (Benedetti 2020: 216). Yet such perceptual features are highly contingent: a drug might just as well have been produced with a different color, shape, or flavor. In that sense, the conditioned response could just as easily have been triggered by an alternative feature. For example, Giang et al. (1996) could easily have paired cyclophosphamide with a flavor other than anise, in which case it would have been that alternative flavor that would have produced lymphocyte count decrease. Indeed, pairing cyclophosphamide with a certain color or shape instead of flavor would likely have worked just as well.

Thus, the very mechanisms of placebo effects give us strong reason to think that when a placebo treatment elicits such an effect, many other treatments could just as readily have done so. In other words, the specific type of treatment administered does not make a difference to the patient's response. Note that nevertheless, in an instance of a placebo effect, the treatment still counts as a cause of the response according to interventionism's minimal condition on causation. For that condition only requires that there be *some* intervention on the cause that would change the effect, which is the case here, since the response would not have occurred if the patient had not been administered any treatment at all. But non-specificity of cause means that many interventions on the treatment would not make a difference to the effect: namely, any intervention that replaced the specific treatment administered with some other treatment.[7]

Moreover, when we are dealing with a placebo response, the treatment also lacks specificity of effect. Because placebo effects depend on patient expectations or conditioning history, a treatment that induces a certain response via the placebo effect can easily produce a different response if expectations or conditioned dispositions are different. For example, depending on patient expectations the administration of a placebo pill can affect conditions as

[7] As we will see in §3.3, this fact implies that that placebo effects generally lack 'proportionality', which plays an important role in the second reading of 'non-specificity' to be discussed below.

diverse as pain (McGlashan *et al*. 1969), sexual dysfunction (Bradford & Meston 2011), migraines (El-Chammas *et al*. 2013), cancer-related fatigue (Hoenemeyer *et al*. 2018), and many others. The same point holds for placebo effects due to conditioning: for example, in Giang *et al*.'s study, the experimenters could easily have induced a different response by pairing anise flavor with a treatment for a different disease. A particularly striking feature of this lack of specificity of effect is that placebo responses can reverse direction entirely depending on patient expectations or conditioning history (Stewart-Williams & Podd 2004: 325). For example, depending on expectations the very same placebo pill can have opposite effects on gastric motility (Meissner 2009), and the same placebo inhaler can either decrease or increase airway narrowing depending on what asthmatic patients are told about its effects (Butler & Steptoe 1986). A single placebo can also produce two opposite conditioned responses depending on the unconditioned stimulus with which it has been associated. For example, blue placebo pills generally produce sedation, but have stimulating effects for Italian men, probably because the Italian soccer team wears blue jerseys (Cattaneo *et al*. 1970).

I am not the first to interpret Shapiro's thesis in terms of a lack of one-one specificity. Kirsch (2005) offers a similar reading but rejects it as empirically untenable. His central objection is straightforward: many active treatments also lack one-one specificity, whether of effect or of cause. Prednisone, for instance, is used to treat a wide range of conditions, including arthritis, autoimmune diseases, and certain allergies. Likewise, a single therapeutic response may be produced by diverse active treatments: acetaminophen, morphine, aspirin, and anti-inflammatories all relieve pain. But it would be absurd to classify these effects as merely placebic. Thus, Kirsch concludes, lack of one-one specificity cannot be what distinguishes placebo from non-placebo effects.

There are, however, two replies to this concern. Together, they clarify and strengthen this first interpretation of Shapiro's thesis, showing that it offers a more persuasive account of placebo effects than Kirsch allows. The first reply relies on the fact that the non-specificity of a causal relationship is a matter of degree – a causal relationship can deviate more or less from the ideal of one-one specificity depending on how many alternative causes could have produced the effect, and how many alternative effects the cause could have produced. And while it is certainly true that some active treatment-response relationships lack specificity[8], placebo effects are distinguished by their *extreme* lack of specificity. This is especially evident if we focus on their lack of specificity of cause. As noted above, when a treatment such as a sugar pill induces a response via the placebo effect, countless other treatments could just as well have produced the same effect. Key evidence for that claim comes from further work by Shapiro. In *The Powerful Placebo*, co-authored with his wife the psychiatrist Elaine Shapiro, they document the extraordinary variety of substances once thought to be medical treatments and used as such, including blood, fat, bones, urine and excrement of many animals, molds, honey, vinegar, and many other ingredients (Shapiro & Shapiro, 1997). As they note, those treatments were likely thought to be efficacious in part because they induced placebo effects. And while not all of them were used to treat pain, each *could* have. (If a sugar pill can induce analgesia, why not pig blood, or vinegar?) The lesson is that virtually anything can serve as a placebo painkiller, provided the right expectations are in place. But the lack of specificity in active treatments is much more constrained. For example, while the list of active painkillers is large, it is still limited. By contrast, the list of placebo painkillers likely includes an indefinite number of substances and procedures.

Note that while this response relies on the fact that there are degrees of non-specificity, it does not imply that the distinction between placebos and active treatments is itself a matter of degree. If it had this implication, it would thereby be in tension with medical theory and practice, which assumes a binary distinction between active and placebo treatments – a treatment has to be either one or the other (see Baron et al., 2023; Jylkkä & Hupli, 2024). But the response just proposed does posit a sharp cutoff between placebic and active effects, and hence also between placebos and active treatments: placebo effects are distinguished by the fact that they display an extreme lack of specificity, in the sense that if a treatment produces a certain response via the placebo effect then the list of other treatments that could have produced the same response is virtually endless.

This claim about the extreme lack of specificity of placebo effects invites an objection, though. Suppose a contemporary Western patient experiences pain relief after taking a sugar pill presented by their doctor as a powerful analgesic. Could drinking pig blood *really* have produced

[8] Though some are likely fully one-one specific: for example, as far as we know, the only treatment for hereditary tyrosinemia type 1 (a genetic disorder that consists in an inability to break down tyrosine) is the molecule nitisinone, and hereditary tyrosinemia type 1 is the only disease that nitisinone can cure.

the same outcome? That would require the patient to believe pig blood is a painkiller– an implausible scenario. One might doubt that such far-fetched possibilities should count in assessing specificity. A comparison with Woodward's (2006, 2010) notion of causal stability is helpful here. Stability concerns whether a causal relationship holds under changes in background conditions, but in assessing it we typically only consider scenarios that "do not depart too much from the actual state of affairs or that do not seem too far-fetched" (Woodward 2006: 11). Presumably the same restriction applies when we ask whether some other cause could have produced the effect, or whether the same cause could have produced a different effect.

My reply is that the pig-blood scenario is not as far-fetched as it seems. In clinical contexts, medical practitioners have considerable power over patients' expectations. A doctor could easily have induced the relevant expectations in their patient, e.g. by convincing them that pig blood contains natural analgesics or (perhaps more plausibly) by presenting it as a novel drinkable painkiller. (Here it is worth noting that what the caregiver says to their patient is a voluntary action, and that as Woodward (2006: 33) observes, we tend to treat scenarios resulting from voluntary actions as relevantly close to actuality.) Moreover, beliefs in the therapeutic efficacy of a certain substance or procedure can also arise due to spontaneous remission after treatment exposure (Benedetti 2020: 178) or by observing somebody else display the response to the treatment (Colloca & Benedetti 2009). Though it is unlikely that either of those mechanisms could lead a contemporary Westerner to believe that pig blood relieves pain, they could easily lead them to come to believe in the efficacy of many other more familiar substances.[9]

[9] These remarks help address another one of Kirsch's objections to Shapiro, namely that placebo effects can be quite specific. For example, because it is widely known that injections are more effective than pills, patients also tend to react more strongly to *placebo* injections than to placebo pills (de Craen *et al*. 2000). According to Kirsch, there is no difference in specificity between the effect of a placebo injection and the active effect of a 'real' injection: the only difference is that the former are "psychologically mediated" (Kirsch, 2005: 797). However, that very fact does induce a difference in specificity. In the case of placebo injections, it would be relatively easy for a placebo pill to actually produce the same effect: all it would take is for the patient to be convinced (e.g.) by their practitioner that pills are actually as effective as injections. Such a variation in belief does not seem especially far-fetched, and at any rate considerably less so than the changes in human anatomy and physiology that would be needed for injections to have smaller *active* effects than pills.

There is a second, and complementary, response to make to Kirsch's concern that causal relationships between active treatments and therapeutic responses can be non-specific too. Clearly, the degree of specificity of a causal relationship depends on how we carve up the range of alternative causes and effects (Woodward 2010, p. 312). And when a relationship is found to be highly non-specific, a possible reaction is to reconsider the carving of the causal space on which that assessment is based and seek another one that restores specificity (see Blanchard, 2022b, 17). Thus, if we find that a factor E is caused by many seemingly disparate factors, we may be prompted to look for a common property of those factors that explains why they all produce E. Reformulating the causal relationship in terms of that common property then helps regain (some) specificity. Now, in the case of active treatments, that strategy is generally successful: if many different active treatments all produce the same effect, we can typically find common features that explain why they produce the same response. For example, though many seemingly disparate substances can relieve pain, most of them (with some exceptions like acetaminophen and ziconotide) are either anti-inflammatories or opioids. Likewise, the large variety of effects that prednisone produces becomes far less bewildering when we note that they all involve inflammation reduction. Thus, in those cases, the relevant relationships between treatment and response can be redescribed in such a way that one-one specificity is restored or considerably improved. These remarks reinforce the idea that there is a sharp distinction between placebo and active treatments, with active treatments generally within a short distance of the ideal of one-one specificity (at least when individuated in the right way), and placebo treatments falling at the other end of the spectrum.

Now, against this second response, one may object that placebo treatments also lend themselves to a similar procedure of redescription. After all, the various substances and procedures that can produce placebo analgesia all share one of two properties: either they are expected to relieve pain, or associated by conditioning with an active painkiller. When their cause is described in terms of these categories, placebo analgesic effects have high specificity of cause.[10] However, a crucial point here is that the only relevant common properties – being the object of certain expectations, or having a certain conditioning history - are *extrinsic* ones. And a basic principle of modern medicine (I suggest) is that treatments are individuated in terms of

[10] Likewise, the heterogeneous character of the responses that a single placebo can produce can be reduced by reclassifying them under a common category such as 'expected response'.

their intrinsic properties, or dispositional properties supervening on those intrinsic properties. For example, prednisone is defined by its molecular formula, and statins by their ability to inhibit the HMG-CoA reductase enzyme, which supervenes on statins' chemical structure. Likewise, in modern medicine, responses to treatments are individuated in terms of properties that supervene on intrinsic physical properties of the patient's body. (Blood pressure supervenes on the physical state of the circulatory system; knee loss of function supervenes on the physical state of the patient's knee; etc.) These principles allow for a variety of more or less fine-grained individuation schemes for treatments and responses. But on none of them do placebo effects count as specific, as the only properties that their relata have in common are extrinsic ones. By contrast, the properties that (e.g.) the various types of active painkillers share are of the sort licensed by modern medicine's principles of individuation. Opioids are defined by their ability to bind to opioid receptors, and anti-inflammatories by their ability to reduce inflammation; in both cases the relevant dispositions supervene on chemical structure. The claim that placebo effects are massively more non-specific than active effects is therefore a highly robust one, in the sense that it holds given any scheme for individuating treatments and responses consistent with modern medicine.

In short, Shapiro's non-specificity thesis can be successfully defended against Kirsch's concern when spelled out in the following way: *on prevalent medical principles for individuating treatments and therapeutic responses, placebo effects are distinguished by their extreme lack of one-one specificity*.

3. Treatments as mere background conditions of placebo responses

I now turn to a second – though related – sense in which placebo effects can rightly be called 'non-specific'. On this second interpretation, the non-specificity of placebo effects is tied to causal selection, the fact that among all factors that are causally relevant to an effect in a minimal sense, we regard some as "real causes" and others as mere "background conditions". Recall Shapiro's claim that "specific activity is the therapeutic influence attributable solely to the contents or processes of the therapies rendered" (Shapiro & Morris, 1978: 372). One way to read this claim is as implying that in a placebo effect, patient response is not explained solely or even primarily by the treatment: instead, the main cause is another factor, namely the patient's expectations or conditioning history. So understood, Shapiro's thesis is similar to the view of

Stewart-Williams and Podd (2004), according to whom the defining mark of placebos is that "they have no *inherent powers* to produce a given effect"; instead, "the cause [of the response] must be in the recipient" (2004: 325). Again, this should not be read as implying that in a placebo effect, the treatment is not causally relevant to the response at all. Since the response would not have occurred had the treatment not been administered, the treatment-response relationship passes a standard counterfactual dependence test for a minimal form of causation. Rather, the point is that the treatment functions only as a mere background condition. This is similar to the way in which we single out the tossing of the cigarette as the real cause and the presence of oxygen as mere background condition of the fire, though both were necessary for the outcome.

An immediate concern with Shapiro's thesis so understood is that this is too flimsy a basis for defining placebo effects. Many philosophers follow J. S. Mill in holding that our practices of causal selection are 'capricious' and 'without any scientific ground' (Mill, 1843: 238). If this is right, the proposal under consideration cannot yield a principled and objective ground for distinguishing placebos from active treatments.

We can address this concern by drawing once again on the interventionist account of causation. In the same paper where he introduces his two notions of specificity, Woodward (2010) also identifies two further dimensions (proportionality and stability) along which relationships that satisfy the minimal interventionist condition on causation can vary. He argues that variations along dimensions of specificity, proportionality and stability supply a principled basis for judgments of causal importance.[11] On the interventionist view of causation, the point of causal selection judgments is to identify variables that afford reliable control over their effects. As Woodward further shows, on that hypothesis we should expect those judgments to favor causal variables that score high on dimensions of specificity, proportionality and/or stability, as such variables tend to be better handles to manipulate their effects. As we will now see, this interventionist account of causal selection supplies principled reasons to regard treatments as mere background conditions of responses produced via the placebo effect, as in such a case the causal relationship between treatment and response fares poorly along these objective dimensions. This strengthens and validates the second reading of Shapiro's non-specificity thesis. Moreover, this interpretation connects closely with the first: one reason treatments count

[11] See Lean (2020) and Ross (forthcoming) for developments of this account of causal selection.

as mere background conditions of placebo responses is precisely the marked lack of one-one specificity of the relationship. In the remainder of this section, I examine how placebo effects' lack of specificity of effect (3.1), lack of specificity of cause (3.2) and lack of stability (3.3) all contribute to making placebos secondary causes of their effects, before outlining some key conceptual benefits of this second version of Shapiro's non-specificity thesis (3.4).

3.1. Lack of specificity of effect

Consider first the lack of specificity of effect. This means, remember, that a treatment producing a response via the placebo effect can easily produce different and even opposite therapeutic responses from one occasion or patient to the next. Intuitively, this implies that in an instance of the placebo effect, the treatment by itself does little to explain patient response. Because it is not uniquely associated with a specific response, the treatment only explains why some response occurred rather than none at all; it is the patient expectations or conditioned dispositions that explain the specific features of the response such as its bodily location, nature, and intensity. This is one principled way in which patient expectations and conditioning history are more important causes of responses produced via placebo effects than the treatments themselves. The interventionist approach to causal selection validates this judgment. When a causal factor C is non-specifically associated with an effect E, and could easily produce different or opposite effects, C by itself provides poor control over the effect. It is no wonder that in those instances we tend to regard C as a mere background condition for E, especially if some further causal factor offers a much better handle of control over precise features of E.

In the case of active effects, the situation is reversed: it is generally the treatment being administered that is specifically associated with the precise response produced, whereas features of the patient that are also involved in producing the effect are not specifically associated with a precise response. For example, the action of a drug depends on various physiological facts about the patient such as blood circulating through their body. But by itself blood circulation can lead to many different therapeutic responses, depending on which substances are present in the bloodstream, and therefore lacks specificity of effect. Here it is the content of the drug that fixes the nature and intensity of the response observed. For that reason (and others discussed below), in this case it is appropriate to regard the treatment as the real cause of the response, and to treat other causally relevant facts about the patient as mere background conditions.

3.2. Lack of specificity of cause

Consider now the fact that placebo effects lack specificity of cause – that is, if a treatment produces a response via the placebo effect, countless other treatments could have induced the same response. The key point here is that specificity of cause is intimately tied to another causal dimension – proportionality – that is important for causal selection. Proportionality is satisfied when the cause's description includes just enough information to account for the effect, but no more than that (Yablo, 1992; Woodward, 2010). For example, the fact that John smoked is enough to explain why he got lung cancer, whereas the fact that he smoked Marlboros contains irrelevant information. As this example shows, our practices of causal selection tend to favor proportional causes: to explain why John got lung cancer, it is more appropriate to cite his smoking than his smoking Marlboros. As a number of authors have argued (List & Menzies, 2009; Woodward, 2010; Blanchard, 2020), interventionism helps make sense of this privilege given to proportional causes. Explanations based on non-proportional causes tend to have misleading implications about the pattern of dependence of the effect on the cause. For instance, claiming that John got lung cancer because he smoked Marlboros wrongly suggests that changing John's brand of cigarette would have prevented the disease. If causal explanation aims to correctly identify interventions that provide control over the explanandum, we should therefore expect causal selection to penalize non-proportional causes.

Now consider a patient who experiences analgesia after taking a sugar pill. Non-specificity of cause, remember, implies that the same effect could have been produced by any number of alternative treatments. Thus, the statement that the patient took a *sugar* pill contains detail that is irrelevant to explaining their response. Consequently, that statement exhibits the defects typical of non-proportional explanations. In particular, it misleadingly suggests that replacing the sugar pill with some other substance would have made a difference to the outcome. But as we have seen, the fact that placebo effects lack specificity of cause implies precisely that those alternatives would have produced the same effect.

In general, then, placebo treatments are not proportional causes of their therapeutic effects, at least when these treatments are individuated in terms of their material composition (as they usually are). This is another principled reason to discount them in causal selection, and to treat expectations or conditioned dispositions as the real causes of placebo effects. Indeed, in our example, the only proportional explanation is that the patient received a treatment they expected to be analgesic. To satisfy proportionality, the treatment must therefore be described in a way that emphasizes the causal importance of expectation in producing the response.

3.3. Lack of stability

The third and final causal dimension that is relevant for the second reading of Shapiro's thesis is stability. As noted above, the stability of a causal relationship has to do with the extent to which the cause would continue to produce the effect under various changes in background circumstances, especially changes involving small departures from actuality (Woodward 2006; 2010). The larger the range of changes under which the cause would remain operative, the more stable the causal relationship. There is ample evidence that our causal selection practices favor highly stable causes.[12] From an interventionist point of view, this is not surprising. A stable cause is one that can be relied upon in many circumstances to influence the effect, thereby providing a reliable and systematic handle of control over it.

Stability considerations provide a third reason to regard treatments as background conditions of placebo responses. Given the way that placebo effects work, we know that the administration of the treatment would not have provoked the response if the patient did not have the right expectations or conditioning history. Importantly, such scenarios typically involve only small departures from actuality, which makes the treatment–response relationship highly unstable. This is because the factors shaping expectations and conditioned dispositions are themselves highly contingent. Patient expectations depend on variables such as verbal and non-verbal behavior of medical practitioners, patient emotional state, the brand and dose of treatment, and social observation (Benedetti 2020), while conditioned dispositions are primarily influenced by the perceptual features of active treatments to which the patient has been exposed in the past. And all those factors are eminently changeable: the practitioner could easily have talked or acted

[12] See Icard *et al*. (2017), Vasilyeva *et al*. (2018) and Woodward (2021).

differently, the patient could easily have been in a different mood on treatment day or failed to be exposed to certain active treatments in the past, the treatment could easily have had a different packaging, etc. This means that in typical instances of placebo effects it would not have taken much for administration of the treatment to induce no response at all, so that the relationship between treatment and response is highly unstable.

Comparison with active effects is instructive here. How a patient responds to an active treatment generally depends on various background circumstances that may include patient age, sex, weight, genetic profile, diet, physiological factors such as hormone levels and metabolism, anatomical variations, disease stage, etc. So when an active effect occurs, many counterfactuals of this form hold:

> If this or that biological fact about the patient had been different, the treatment would not have provoked the response

But here the truth of those counterfactuals does not significantly affect the stability of the treatment-response relationship, because its antecedent typically involves a much larger departure from actuality than a mere change in expectations or conditioning history, and one that is often highly far-fetched. Many of these biological facts (e.g. genetic profile) are fixed at conception, and/or are largely irreversible once in place (e.g. anatomical traits, organ function, etc.). Other features such as weight and metabolic profile can or do change over time, but generally only slowly so. This means that any causally plausible way for these traits to have been different would have had to involve a change relatively far away in the past: for example, any credible scenario in which the patient now has a different genome or age would require a change at time of conception.[13] Such alterations to the distant past typically count as large departures from actuality (Lewis, 1979). By contrast, the factors shaping patient expectations at the time of treatment are inherently variable and prone to sudden, relatively unpredictable changes. This makes their alteration look like a far more serious and proximate possibility than changes in the biological factors modulating active treatment effects.

[13] The notion of causal plausibility comes from Bennett (2003), who emphasizes its importance in counterfactual evaluation.

These considerations underscore a key difference between placebo and active treatments. If placebo effects are highly sensitive to fluctuating circumstances, we should expect the efficacy of placebo treatments to have low between- and even within-subject reproducibility. Empirical evidence supports that prediction. For example, an influential study of placebo analgesia in pregnant women (Liberman, 1967) found that response in one situation did not predict response in other situations. In a more recent study, participants were given a placebo analgesic cream on four occasions. Response on one occasion significantly predicted response on others only when the placebos bore the same name, suggesting that "even very small changes in the context of presentation can affect individual differences in the placebo response" (Whalley et al., 2008, p. 540). Other studies have concluded that "the amount of relief obtained with placebo varies considerably between patients" (McQuay *et al*. 1996), that placebo responses display "large within-subject variability" ((Horing *et al*. 2014: 2), and that "when looking at a particular patient, placebo effects are unpredictable and hard to reproduce" (Arandia & Di Paolo 2021).

By contrast, because the features that modulate the efficacy of active treatments tend to be relatively fixed, we can expect the efficacy of active treatments to exhibit greater consistency. This remains true even when the relevant background conditions for efficacy are uncommon or time-limited (e.g., only until the disease reaches its next stage). For even then one can still expect the treatment to be reliably efficacious for the patient during that period of time. This also means that even if an active treatment only works in statistically unlikely circumstances, one may still be able to formulate restricted causal statements that have a very high degree of stability (e.g. "This treatment works well for you" or "The chemotherapy should remain effective as long as the cancer does not metastasize.") By contrast, the inherent variability in the factors involved in placebo effects plausibly means that even such highly restricted causal statements fail to hold for placebo effects.

As we saw in §§3.1–3.2, considerations of specificity and proportionality not only penalize treatments as mere background conditions of placebo responses, but also favor expectations and conditioned dispositions as the "real causes" of these responses. Stability likewise privileges expectations, since these are generally more stable causes of placebo responses than the treatments themselves. Suppose again that a patient's pain decreases after taking a sugar pill presented as a powerful analgesic. Expectations of efficacy would still have produced analgesia had the patient been given any other treatment framed in the same way. In other words, the expectation–response relationship is insensitive to the treatment's material details. (The fact that expectations are stable causes is therefore closely linked to placebo effects' lack of specificity of cause.)

Interestingly, this argument does not apply as straightforwardly to placebo effects driven by conditioning. Consider Giang et al.'s (1996) study, in which decreased lymphocyte count was induced by anise syrup. Here the effect seems to depend quite sensitively on the treatment: the conditioned disposition would not have produced the response if the patient had ingested a syrup with a different flavor, for example. In such cases, stability considerations do not clearly favor the conditioned disposition over the treatment. Still, as we saw earlier, there are other principled reasons to regard conditioned dispositions as more important causes than the treatment itself, though in conditioning-based placebo effects the difference is perhaps less stark than in expectation-driven cases.

3.4. Placebos as second-rate treatments and Beecher's paradox

In sum, the interventionist account of causal dimensions and causal selection provides strong support for the second reading of Shapiro's thesis. It identifies several objective dimensions that render treatments mere background conditions of the responses they provoke via the placebo effect, thereby supplying a principled basis for distinguishing placebo effects from active effects. Moreover, the interventionist account nicely explains the *point* of regarding treatments as mere background conditions of placebo responses. Consider in particular the fact that placebo effects lack stability and specificity of effect. A cause that does not produce an effect stably and/or specifically is one that cannot be reliably manipulated to produce that precise effect and thus provides poor control over it. This means that if a treatment produces a response through a placebo effect, the administration of the treatment alone is not a reliable way to produce the response. Instead, to reliably induce the response, one must also ensure that the patient's expectations and/or conditioned dispositions – the real *loci* of control over the response – are appropriately configured. If one wants to induce an effect via conditioning, one first has to ensure that the patient has been subjected to the right conditioning protocol. For expectancy-based effects, the practitioner may have to engage in careful management: adopting the right verbal and non-verbal behavior, ensuring that the treatment is delivered in plain sight, and maintaining the integrity of the "healing ritual." The practitioner may also have to carefully monitor the patient to preserve their expectations of efficacy (e.g., by preventing them from discovering that they are given a placebo). Finally, since placebo effects also depend on highly changeable factors over which the practitioner has little control, such as patient mood, the success of a therapeutic strategy relying on the placebo effect may heavily depend on luck. And indeed, as we saw above, there is strong evidence that placebo effects lack reproducibility and consistency.[14] True, as also noted, whether an active treatment produces its intended active effect in a patient also typically depends on factors beyond the caregiver's control. But many of the relevant factors (age, sex, physiological parameters, etc.) can be ascertained before treatment, often making it possible to assess in advance whether a treatment is likely to work for a given patient. Even in cases where this cannot be ascertained in advance, the relative fixity of the relevant background factors often means that if a treatment is found to be effective in a given individual on a certain occasion, one can expect it to continue to work on other occasions also, at least for a certain amount of time.

These remarks point to a significant conceptual benefit of Shapiro's thesis (on this second reading): it helps make sense of the prevalence of negative attitudes toward placebos, which is well-documented among both medical practitioners and laypeople (Bernstein et al., 2020; Bishop et al. 2014). Witness the fact that placebo therapies are often described as 'sham', or the negative implications of the common claim that the treatments of alternative medicine are 'just placebos'. Because placebos do have real effects, some authors (e.g. Gøtzsche, 1994) take the prevalence of those attitudes as a reason to regard the placebo/active treatment distinction as unhelpful. But on the account I proposed, these attitudes are reasonable. Despite recurrent calls to "harness the power of the placebo effect" (Benson & Friedman, 1996), medical practitioners' and laypeople's skepticism toward placebos is not unwarranted, and reflects causal differences between placebo and active treatments that matter for therapeutic practice.

Besides explaining why placebos are treated as second-rate treatments, the second reading of Shapiro's thesis has another significant conceptual benefit: it helps make sense of a paradox at the heart of the notion of placebo effect. This is the fact that the notion implies that

[14] One finds some recognition in the literature that placebo effects' lack of stability is a significant obstacle to their therapeutic use. A 2010 UK official report on homeopathy, for example, states that "prescribing pure placebos is bad medicine. Their effect is unreliable and unpredictable and cannot form the sole basis of any treatment." (House of Commons Science and Technology Committee, 2010). And a paper tellingly titled " 'Caution, this treatment is a placebo. It might work, but it might not' " (Beedie et al., 2018) cites the considerable between and within-subject variability as a key reason not to prescribe placebos in sport medicine.

placebos cause medical outcomes, and yet "placebo" strongly connotes inertness. We may call this Beecher's paradox, since that conceptual tension is evident in Beecher's (1955) early definition of placebo effects as the "therapeutic effects" of "pharmacologically inert substances". Some eliminativists (e.g. Nunn, 2009) have taken the paradox to show that the notion of placebo effect is fundamentally inconsistent. But Shapiro's thesis, understood in the second sense proposed here, dissolves the threat raised by the paradox. On the one hand, placebos *are* causes of their therapeutic effects in the minimal sense that in some circumstances administering a placebo makes a difference to the medical state of the patient. (Similarly, the presence of oxygen is a cause of the fire in the minimal sense that the latter wouldn't have occurred without the former.) But on the other hand, more discriminative practices of causal selection also tend to select features of the patient as the main (or 'real') causes of placebo effects, relegating the treatments themselves to the status of mere background conditions. This plausibly explains our tendency to see placebos as lacking "inherent causal powers" (Stewart-Williams & Podd 2004) to produce medical responses. Thus, when we take into account that 'cause' can be understood in more or less demanding senses, we see that there is no fatal inconsistency at the core of the notion of placebo effect. That being said, these remarks provide support for the idea that the phrase 'placebo effect' is a misnomer. That idea is commonly voiced in the literature, with researchers proposing to rename the effect "remembered wellness" (Benson & Friedman 1996), "meaning effect" (Moerman 2002) or "contextual healing" (Miller & Kaptchuk 2008). Examining these proposals is beyond the scope of this paper, but on the view proposed here the idea that "placebo effects" are misnamed makes sense, as placebo treatments are not the most important causes of the effects they produce: in this way, speaking of "placebo effects" is like calling fire an "oxygen effect".

4. Comparison with other accounts

To summarize, I have argued that placebo effects are non-specific in two senses. First, they display an extreme lack of one-one specificity, and second, the outcome of a placebo effect cannot be attributed specifically to the treatment: instead, our practices of causal selection favor patient expectations and conditioned dispositions as the true causes of the response. The latter fact is itself due to placebo effects' lack of one-one specificity (and closely connected features such as lack of proportionality and stability); thus, these two senses are intimately related to one another. Together, they provide a plausible interpretation of what Shapiro meant to say, and supply a clear conceptual footing to address the central questions raised by the placebo effect. Indeed, the non-specificity account directly sheds light on some of these central questions, such as the status of alternative medicine and why we tend to regard placebos as inert. To highlight some further benefits of the account presented in this paper, I will close by briefly comparing it with the three leading views of placebo effects mentioned in section 1.

Consider first Kirsch's definition of the placebo effect as "that portion of the treatment effect that was produced psychologically, rather than through physical means" (2005: 791). As noted above, that definition presupposes an untenable form of dualism, and ignores the fact that placebo treatments produce real physiological responses. Neither of these issues arises for Shapiro's thesis. Another problem for Kirsch's view is its implication that all psychotherapies are placebos – a question that should be settled on empirical not definitional grounds (Howick, 2017). Though Shapiro's original (1968) account also had this issue, the non-specificity thesis suitably spelled out does not. Indeed, it supplies a plausible explanation of when and why a psychotherapy is a placebo. To illustrate, suppose that the positive therapeutic effects - if any – of (say) Lacanian psychoanalysis are due entirely to patients' positive expectations about that psychotherapeutic framework or to conditioning. Then patients could easily show no response (or a negative response) to the therapy if they had different expectations or a different conditioning history. Moreover, any other type of psychotherapy could easily produce the same effect on patients, if only they had the right expectations or an appropriate conditioning history. In those circumstances, Shapiro's thesis deems Lacanian psychoanalysis to be placebic. On the other hand, suppose that the effects of Lacanian psychoanalysis are fairly one-one specific (i.e. they are generally the same, and no or few other types of psychotherapy can produce the same ones) and show some stability. (For example, they do not disappear when expectations of treatment efficacy disappear.) Then Shapiro's thesis would rule – plausibly, it seems to me – that Lacanian psychoanalysis is an active treatment.[15]

[15] That being said, the case of psychotherapy raises certain complications having to do with the individuation of psychotherapeutic treatments. These issues are apparent in discussions of the so-called 'Dodo effect', the hypothesis first put forward by Rosenzweig (1936) that all psychotherapies share common factors that make them equally effective. Those common features involve the creation of a "therapeutic alliance" with the therapist built on hope, warmth and expectation of improvement, leading the patient to reconceptualize their problems and change their emotional reaction to them, which in turn leads them to address these problems via behavioral regulation. (See Lambert & Ogles, 2021). Supposing the hypothesis is true (a matter of debate), a further question is whether this would mean

Consider next Grünbaum's (1986) account of the placebo effect as any effect produced by an incidental feature of treatment, i.e. any feature that the theory behind the treatment regards as causally irrelevant to patient response. As we have seen, a major worry for the account is that it remains too uninformative (Friesen 2020), because it makes placebo effects theory-relative and hence does not identify any interesting feature that placebo effects share. The non-specificity thesis, on the other hand, precisely identifies a robust feature that placebo effects have in common. Consequently, its explanatory power is higher than Grünbaum's. For example, as we have seen, the non-specificity thesis explains why placebos are generally treated as second-rate treatments, and why we tend to regard placebos both as inert and as causes of therapeutic effects. By contrast, as far as I can see, Grünbaum's account sheds no light on those issues.

Finally, consider Friesen's proposal to define the placebo effect through the mechanisms that underlie it, viz. as "any positive clinical outcome that is brought about via expectations or conditioning" (2020, 14). Though Friesen allows that her definition may have to be revised if evidence of new mechanisms emerges, her account does not explain when and why it would be

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that the effect of psychotherapies is placebic. Some authors indeed categorize the Dodo effect as a kind of placebo effect (e.g. Hovda, 2019), while others are careful to distinguish them (e.g. Enck & Zipfel, 2019). The non-specificity account can shed light on why there is a controversy here. At first glance, the Dodo effect does indeed look like a case of placebo effect, since it implies that the effect of a given psychotherapy is non-specific – by hypothesis, another psychotherapeutic method would have produced the same outcome. However, matters are in fact not so straightforward. The key point is that the common factors that are hypothesized to be driving the Dodo effect are part of the psychotherapeutic treatment itself: every psychotherapeutic treatment intends to lead the patient to a series of sequential emotional and behavioral changes through the establishment of a certain rapport with the therapist. Seen in this way, the Dodo effect seems to be a case where, in the manner discussed at the end of §2, we can identify a common feature belonging to all relevant treatments that explains why they all produce the same effect, which can therefore be recast as a specific effect. This counts in favor of classifying psychotherapies as active treatments if the Dodo hypothesis is true. (Note that this reaction is not available in cases where the effect of a somatic treatment or drug therapy occurs because of the atmosphere of warmth and expectation of improvement created by the healing context: for in that case the relevant contextual features are extrinsic features of the treatment and *not* part of how it is individuated.) On the other hand, every psychotherapeutic treatment also includes a more specific set of methods and techniques (beyond the general common factors just outlined) that sets it apart from other psychotherapeutic methods. And here the Dodo hypothesis suggests that the effect of those techniques is non-specific – any other method would do just as well. Conceptualized in this way, the Dodo hypothesis provides a reason to regard the effect produced by a particular psychotherapeutic method as non-specific, which explains why there is some inclination to regard the Dodo effect as placebic. Here, however, there is an open question about *how* extreme the non-specificity is. Suppose a patient experiences psychological improvement after undergoing Cognitive Behavioral Therapy. Would the effect still have occurred for any technique presented by the therapist as an effective way for the patient to work through their problems, say watching a film or discussing current events? As far as I know, there is little empirical evidence bearing on that issue. The activities just mentioned are standard placebo controls for evaluating the effectiveness of psychotherapies (Prioleau et al., 1983), but in the relevant experiments they are *not* presented to the patient as effective psychotherapies and do not involve building a therapeutic alliance with a therapist. This makes the results of the relevant studies irrelevant to the question of whether the evaluated psychotherapeutic method has a *specific* effect on recovery that is not merely due to common factors shared with other therapies.

legitimate to regard a newly discovered therapeutic effect as a kind of placebo effect. Shapiro's thesis supplies a straightforward answer: any mechanism that legitimately counts as a placebo mechanism should be one that produces effects displaying the characteristic non-specificity pattern identified in this paper and that we observe in effects produced by expectations and conditioning. In this way, while the non-specificity account does not define placebos in terms of mechanisms, it is compatible with and even complementary to mechanistic inquiry, and may in fact serve as a useful guide to identifying further candidate placebo mechanisms. But this complementarity is possible precisely because the non-specificity account operates at a different level from mechanistic inquiry, reflecting the fact that the definitional question of what makes something a placebo effect is distinct from the explanatory question of how placebo effects work. The latter question is a fascinating one for empirical research, but answering it is not required in order to characterize what placebo effects are and what sets them apart from active effects. Indeed, Friesen's proposal illustrates the risks of conflating these two questions: by defining placebo effects in terms of their currently known mechanisms, her account is unable to explain what expectancy and conditioning have in common such that it makes sense to group them in a single medical category, and to separate them from active effects. This gap leaves eliminativist worries about the usefulness of the notion insufficiently addressed. Because it is framed at the level of causal structure rather than mechanism, the non-specificity thesis is able to provide a clear answer. What effects driven by expectancy and conditioning share is their non-specificity, understood in the two related ways proposed in this paper. That feature sets these effects apart from other medical responses in therapeutically important ways, so that the distinction between placebo and non-placebo effects does have a central role to play in medicine.

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